

Cod liver oil intake and incidence of asthma in Norwegian adults--the HUNT study.

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BACKGROUND: Cod liver oil is an important source of vitamin D, but also contains other fat-soluble components such as vitamin A. Before 1999, the cod liver oil formula in Norway contained a high concentration of vitamin A (1000 µg per 5 ml). High vitamin A status is associated with increased risks of several chronic diseases. **OBJECTIVE:** To investigate the association between cod liver oil intake and asthma development. **METHODS:** In the Nord-Trøndelag Health Study, a total of 25 616 Norwegian adults aged 19-55 years were followed up from 1995-1997 to 2006-2008. Current analysis based on 17 528 subjects who were free of asthma and had complete information on cod liver oil intake at baseline. Cod liver oil intake was defined as daily intake ≥ 1 month during the year prior to baseline. Incident asthma was reported as new-onset asthma during the 11-year follow-up. **RESULTS:** Of the 17 528 subjects, 18% (n=3076) consumed cod liver oil daily for ≥ 1 month over the past year. Cod liver oil intake was significantly associated with incident asthma with an OR of 1.62 (95% CI 1.32 to 1.98) after adjustment for age, sex, daily smoking, physical activity, education, socio-economic status, family history of asthma, and body mass index (BMI). The positive association was consistent across age (< 40/ $\geq$ 40 years), sex (men/women), family history of asthma (yes/no) and BMI subgroups (< 25/ $\geq$ 25 kg/m²). **CONCLUSIONS:** Intake of cod liver oil with high vitamin A content was significantly associated with increased incidence of adult-onset asthma.